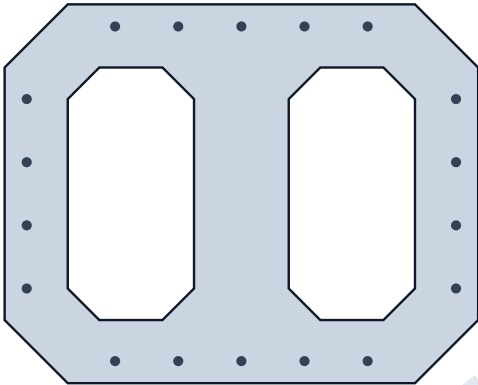


1. Input

Section, materials and resistance basis exactly as analysed.

Project information			
Project Name	PM-advanced (7) 2D reference case		
Client	Reference verification		
Company	P-M Column Designer		
Designed by	Engineering example	Checked by	Independent review
Address	Seoul, Korea	Date	2026-07-23
Calculation identity			
Section	PM-advanced (7) 2D — 1500x1200 .	Calculation profile	KDS 2024 — Stress-strain integratio.
Mechanics	Stress-strain integration	Resistance profile	KDS 2024 current set · resistance: K.
Profile edition	2024 partial-revision framework (effe.	Verification status	draft
Concrete model basis	Code-default	Method id	kds-2024-current-set-global-strength.
Units	mm · N · MPa; forces reported in kN,.	Sign convention	Compression positive; Mx = ΣF·(y-yc.
Section			
Cross-section (mm, about the net concrete centroid)		18 bars	Section properties
			Gross concrete area Ag1,120,000 mm²
			Reinforcement area As14,476 mm²
			Reinforcement ratio ρ1.293 %
			Bar count18
			Net centroid(0.0, 0.0) mm
			Bounding box1,500.0 × 1,200.0 mm
			Solids / holes1 / 2
			Perimeter9,262.7 mm
Materials			
Name	KDS C30 (workbook Input)	Standard	KDS
fck	30.0 MPa	Model	kds-parabolic
εcu	0.00330	εc0	0.00200
α	0.8500	Ec	28,417 MPa
Tension	ignored	Name	SD400 (workbook Input)
Standard	KDS	fy	400.0 MPa
Es	200,000 MPa	Model	elastic-perfectly-plastic
εy	0.002000	εu	not declared
Grades used	1		
Design resistance basis			
Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.700
φ tension	0.850	εt,limit	0.00500
Transition rule	fixed-or-yield-multiple	Axial cap	0.800 × factored compression pole
Sampling and numerical evidence			
Sampling mode	fixed	Active Design points	972
Active Design directions	36	Active Design stations	27
Active Nominal points	972	Active Nominal directions	36
Active Nominal stations	27	Direction tolerance	— %
Worst interpolation error	— %	Tolerance reached	yes
Concrete discretization	864 cells / 5196 points	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

Warning: Design profile status is draft; results are for review, not release.

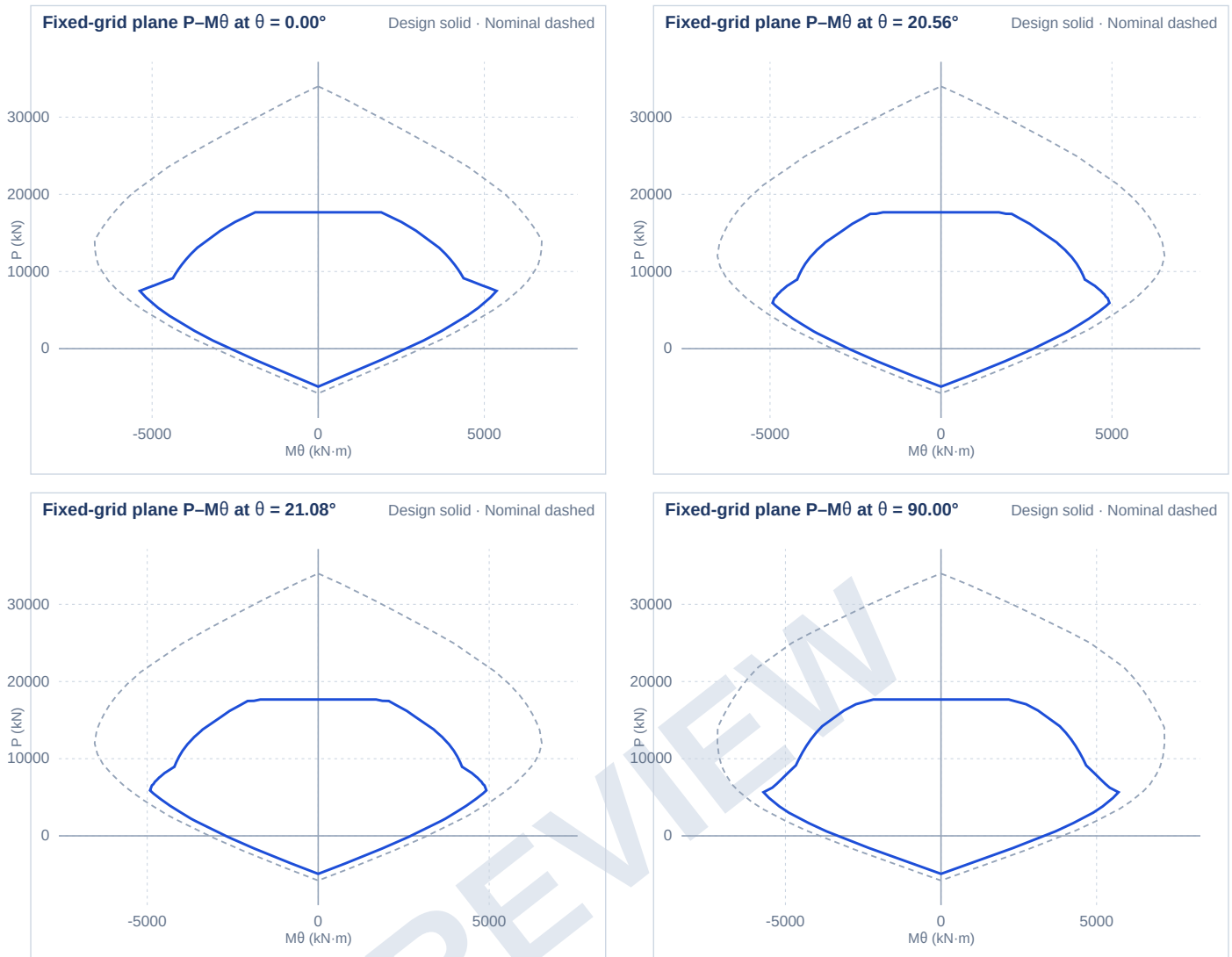
Reinforcement schedule

#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	32.0	-400.0	530.0	804.2	SD400 (workbook Input)
2	2	32.0	-200.0	530.0	804.2	SD400 (workbook Input)
3	3	32.0	0.0	530.0	804.2	SD400 (workbook Input)
4	4	32.0	200.0	530.0	804.2	SD400 (workbook Input)
5	5	32.0	400.0	530.0	804.2	SD400 (workbook Input)
6	6	32.0	-400.0	-530.0	804.2	SD400 (workbook Input)
7	7	32.0	-200.0	-530.0	804.2	SD400 (workbook Input)
8	8	32.0	0.0	-530.0	804.2	SD400 (workbook Input)
9	9	32.0	200.0	-530.0	804.2	SD400 (workbook Input)
10	10	32.0	400.0	-530.0	804.2	SD400 (workbook Input)
11	11	32.0	-680.0	300.0	804.2	SD400 (workbook Input)
12	12	32.0	-680.0	100.0	804.2	SD400 (workbook Input)
13	13	32.0	-680.0	-100.0	804.2	SD400 (workbook Input)
14	14	32.0	-680.0	-300.0	804.2	SD400 (workbook Input)
15	15	32.0	680.0	300.0	804.2	SD400 (workbook Input)
16	16	32.0	680.0	100.0	804.2	SD400 (workbook Input)
17	17	32.0	680.0	-100.0	804.2	SD400 (workbook Input)
18	18	32.0	680.0	-300.0	804.2	SD400 (workbook Input)

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 17670.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	φ	Class	UR	Verdict
1	ULS-1 workbook Newton d.	24942.9	3714.2	1431.8	21.1	0.650	compression-co.	1.539	INADEQUA.
2	ULS-2 same axial, half mo.	24942.9	1857.1	715.9	21.1	0.650	compression-co.	1.412	INADEQUA.
3	ULS-3 low axial, high mom.	5000.0	4000.0	1500.0	20.6	0.850	transition	0.877	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

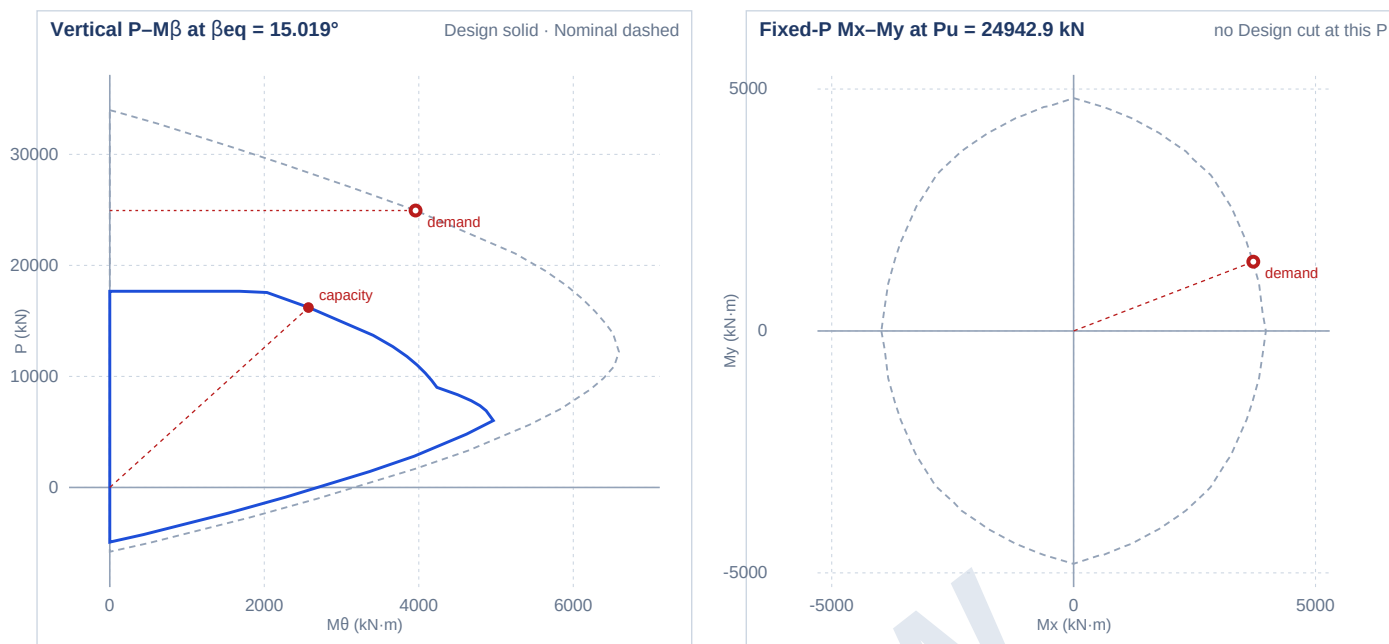
4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.5394 · fixed-P diagnostic —

Converged preview equilibrium inside the material domain.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	24,942.9 kN	M_{ux}	3,714.2 kN·m
M_{uy}	1,431.8 kN·m	Moment direction θ_L	21.08°
Resultant moment	3,980.6 kN·m	Strain-plane ϵ_0	1.5162e-3
Curvature κ_x	2.3759e-6	Curvature κ_y	6.3748e-7
Strain direction β_{eq}	15.019°	Neutral-axis depth c	1,338.4 mm
Capacity at P_u	—	ϕ	0.6500
Classification	compression-controlled	Utilization UR	1.5394
Fixed-P ratio	—	Verdict	INADEQUATE

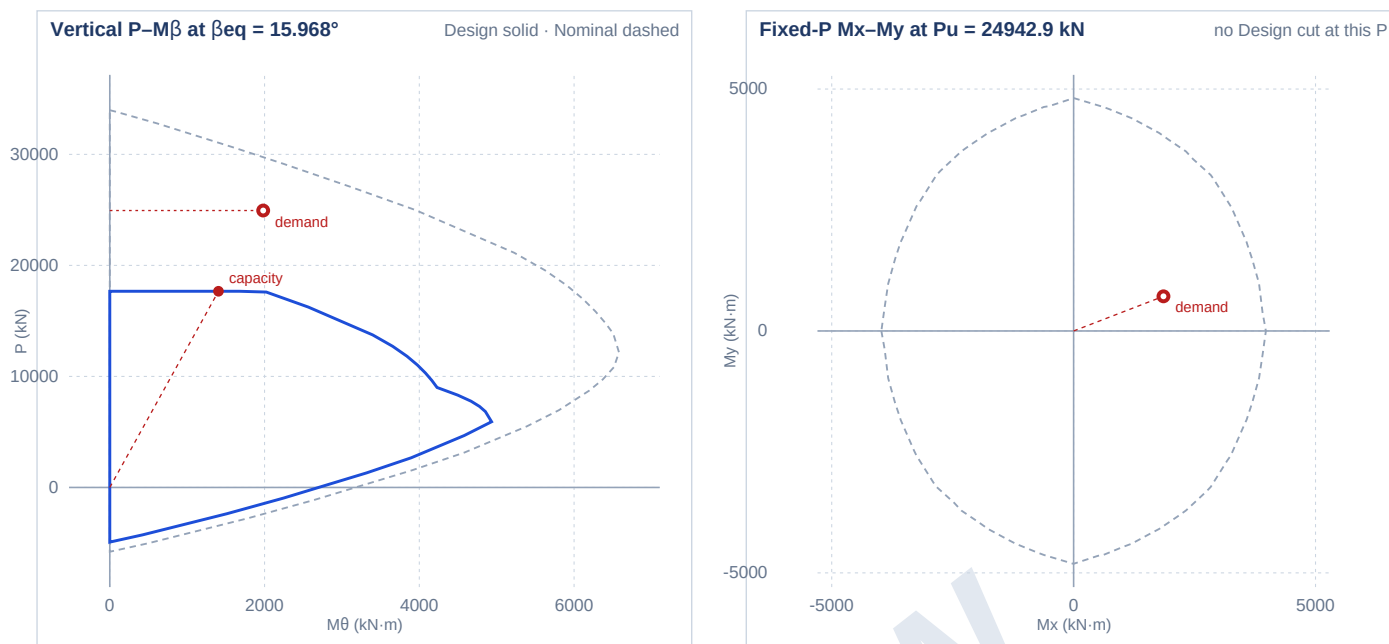
4.2 ULS-2 same axial, half moment (inside)

$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$ · $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.4116 · fixed-P diagnostic —

Converged preview equilibrium inside the material domain.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	24,942.9 kN	M_{ux}	1,857.1 kN·m
M_{uy}	715.9 kN·m	Moment direction θ_L	21.08°
Resultant moment	1,990.3 kN·m	Strain-plane ϵ_0	1.0984e-3
Curvature κ_x	7.8173e-7	Curvature κ_y	2.2369e-7
Strain direction β_{eq}	15.968°	Neutral-axis depth c	2,079.0 mm
Capacity at P_u	—	ϕ	0.6500
Classification	compression-controlled	Utilization UR	1.4116
Fixed-P ratio	—	Verdict	INADEQUATE

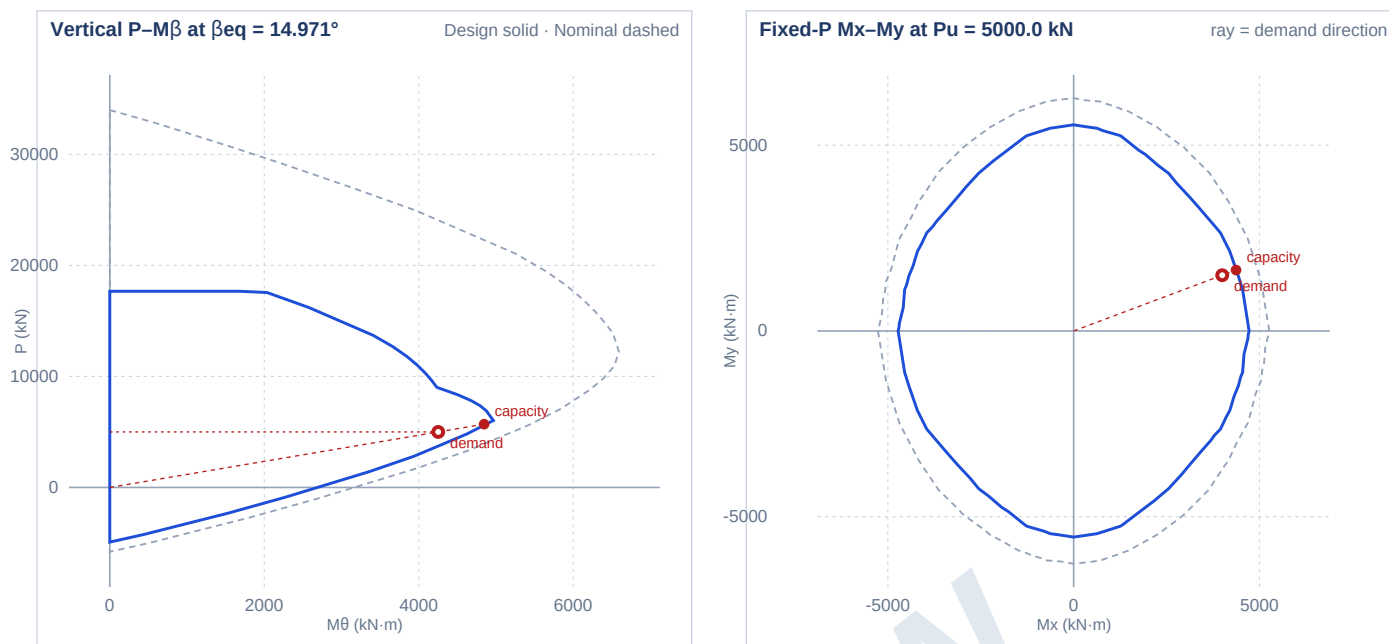
4.3 ULS-3 low axial, high moment

$P_u = 5000.0 \text{ kN}$ · $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 20.56^\circ$

ADEQUATE · utilization **0.8774** · fixed-P diagnostic **0.9138**

Converged preview equilibrium inside the material domain.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

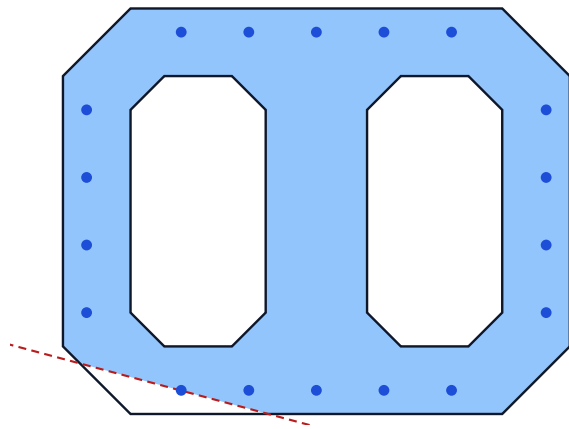
P_u	5,000.0 kN	M_{ux}	4,000.0 kN·m
M_{uy}	1,500.0 kN·m	Moment direction θ_L	20.56°
Resultant moment	4,272.0 kN·m	Strain-plane ϵ_0	-3.8060e-4
Curvature κ_x	2.5504e-6	Curvature κ_y	6.8197e-7
Strain direction β_{eq}	14.971°	Neutral-axis depth c	577.5 mm
Capacity at P_u	$M_x 4,372.1 \cdot M_y 1,639.5 \text{ kN}\cdot\text{m}$	ϕ	0.8500
Classification	transition	Utilization UR	0.8774
Fixed-P ratio	0.9138	Verdict	ADEQUATE

5.1 ULS-1 workbook Newton demand (on the 15 deg surface)

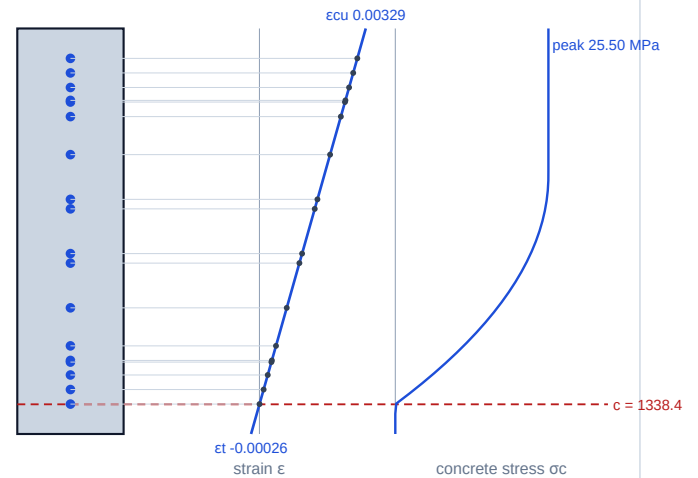
Worked section state and the points behind both curves. $P_u = 24942.9 \text{ kN}$ · $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta=1444$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Concrete force	21,577.4 kN	Concrete Mx	2,900.4 kN·m
Concrete My	1,122.3 kN·m	Steel force (gross)	3,629.5 kN
Displaced concrete	-264.0 kN	Steel net	3,365.5 kN
Response P	24,942.9 kN	Response Mx	3,714.2 kN·m
Response My	1,431.8 kN·m	Residual $ \Delta P $	1.01e-4 N
Residual $ \Delta M_x $	3.15e-3 N·mm	Residual $ \Delta M_y $	4.99e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.8	2.520e-3	400.00	321.70	—
2	-200.0	530.0	804.2	262.0	2.648e-3	400.00	321.70	—
3	0.0	530.0	804.2	210.1	2.775e-3	400.00	321.70	—
4	200.0	530.0	804.2	158.3	2.903e-3	400.00	321.70	—
5	400.0	530.0	804.2	106.5	3.030e-3	400.00	321.70	—
6	-400.0	-530.0	804.2	1337.6	1.971e-6	0.39	0.32	—
7	-200.0	-530.0	804.2	1285.8	1.295e-4	25.89	20.82	—
8	0.0	-530.0	804.2	1233.9	2.570e-4	51.39	41.33	—
9	200.0	-530.0	804.2	1182.1	3.845e-4	76.89	61.84	—
10	400.0	-530.0	804.2	1130.3	5.120e-4	102.39	82.35	—
11	-680.0	300.0	804.2	608.5	1.795e-3	359.09	288.80	—
12	-680.0	100.0	804.2	801.7	1.320e-3	264.06	212.37	—
13	-680.0	-100.0	804.2	994.8	8.451e-4	169.02	135.94	—
14	-680.0	-300.0	804.2	1188.0	3.699e-4	73.99	59.50	—
15	680.0	300.0	804.2	256.1	2.662e-3	400.00	321.70	—
16	680.0	100.0	804.2	449.2	2.187e-3	400.00	321.70	—
17	680.0	-100.0	804.2	642.4	1.712e-3	342.42	275.39	—
18	680.0	-300.0	804.2	835.6	1.237e-3	247.38	198.96	—

Solver and admissibility evidence

Strain plane ε_0	1.5162e-3	Curvature k_x	2.3759e-6
Curvature k_y	6.3748e-7	Neutral-axis line	164.9806°
Compression depth c	1,338.387 mm	Iterations	7
Relative residual	2.576e-12	Max concrete compression	0.003292
Concrete limit	0.003300	Max steel tension	0.000000

Steel limit

not declared

Strain admissibility

admissible

Governing utilization

1.5394

Vertical curve points — direct meridian at $\beta_{eq} = 15.019^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN-m)
1	Design	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	17,670.3	0.00
2	Design	station-1	$c/D = 3$	2.7500e-3	7.3572e-7	1.9740e-7	17,670.3	335.97
3	Design	station-2	$c/D = 2$	2.4750e-3	1.1036e-6	2.9610e-7	17,670.3	671.95
4	Design	station-3	$c/D = 1.5$	2.2000e-3	1.4714e-6	3.9480e-7	17,670.3	1,007.92
5	Design	station-4	$c/D = 1.2$	1.9250e-3	1.8393e-6	4.9350e-7	17,670.3	1,343.90
6	Design	station-5	$c/D = 1.1$	1.8000e-3	2.0065e-6	5.3837e-7	17,670.3	1,679.87
7	Design	station-6	$c/D = 1$	1.6500e-3	2.2071e-6	5.9220e-7	17,558.6	2,034.89
8	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.5186e-3	2.3828e-6	6.3935e-7	16,212.6	2,574.90
9	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2487e-3	2.7439e-6	7.3622e-7	13,702.1	3,409.27
10	Design	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1138e-3	2.9244e-6	7.8465e-7	12,683.1	3,664.28
11	Design	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7885e-4	3.1049e-6	8.3309e-7	11,800.1	3,846.28
12	Design	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4390e-4	3.2854e-6	8.8152e-7	11,020.8	3,979.53
13	Design	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0894e-4	3.4660e-6	9.2996e-7	10,304.6	4,083.62
14	Design	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7399e-4	3.6465e-6	9.7840e-7	9,639.9	4,166.53
15	Design	station-14	$\epsilon_s/\epsilon_y = 1$	4.3904e-4	3.8270e-6	1.0268e-6	9,016.4	4,234.01
16	Design	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6914e-4	4.1880e-6	1.1237e-6	8,348.9	4,504.03
17	Design	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0076e-4	4.5491e-6	1.2206e-6	7,814.9	4,678.59
18	Design	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7066e-4	4.9101e-6	1.3174e-6	7,366.3	4,792.00
19	Design	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4056e-4	5.2712e-6	1.4143e-6	6,910.3	4,871.99
20	Design	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1804e-3	5.9932e-6	1.6081e-6	6,025.0	4,964.47
21	Design	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7202e-3	6.7153e-6	1.8018e-6	4,802.8	4,618.49
22	Design	station-21	$\epsilon_s/\epsilon_y = 4$	-2.7998e-3	8.1595e-6	2.1893e-6	2,819.9	3,942.47
23	Design	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8794e-3	9.6036e-6	2.5768e-6	1,437.4	3,365.61
24	Design	station-23	$\epsilon_s/\epsilon_y = 7.5$	-6.5784e-3	1.3214e-5	3.5455e-6	-856.6	2,284.53
25	Design	station-24	$\epsilon_s/\epsilon_y = 10$	-9.2774e-3	1.6824e-5	4.5142e-6	-2,301.4	1,539.66
26	Design	station-25	$\epsilon_s/\epsilon_y = 20$	-2.0073e-2	3.1266e-5	8.3890e-6	-4,240.2	438.24
27	Design	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-4,922.0	0.00
1	Nominal	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	33,981.4	-0.00
2	Nominal	station-1	$c/D = 3$	2.7500e-3	7.3572e-7	1.9740e-7	33,981.4	-0.00
3	Nominal	station-2	$c/D = 2$	2.4750e-3	1.1036e-6	2.9610e-7	33,850.6	74.25
4	Nominal	station-3	$c/D = 1.5$	2.2000e-3	1.4714e-6	3.9480e-7	32,775.0	616.42
5	Nominal	station-4	$c/D = 1.2$	1.9250e-3	1.8393e-6	4.9350e-7	30,528.3	1,636.68
6	Nominal	station-5	$c/D = 1.1$	1.8000e-3	2.0065e-6	5.3837e-7	29,098.7	2,258.21
7	Nominal	station-6	$c/D = 1$	1.6500e-3	2.2071e-6	5.9220e-7	27,013.3	3,130.60
8	Nominal	station-7	$\epsilon_s/\epsilon_y = 0$	1.5186e-3	2.3828e-6	6.3935e-7	24,942.5	3,961.38
9	Nominal	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2487e-3	2.7439e-6	7.3622e-7	21,080.2	5,245.04
10	Nominal	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1138e-3	2.9244e-6	7.8465e-7	19,512.4	5,637.36
11	Nominal	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7885e-4	3.1049e-6	8.3309e-7	18,153.9	5,917.35
12	Nominal	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4390e-4	3.2854e-6	8.8152e-7	16,955.1	6,122.35
13	Nominal	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0894e-4	3.4660e-6	9.2996e-7	15,853.3	6,282.50
14	Nominal	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7399e-4	3.6465e-6	9.7840e-7	14,830.7	6,410.04
15	Nominal	station-14	$\epsilon_s/\epsilon_y = 1$	4.3904e-4	3.8270e-6	1.0268e-6	13,871.5	6,513.86
16	Nominal	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6914e-4	4.1880e-6	1.1237e-6	12,217.9	6,591.26
17	Nominal	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0076e-4	4.5491e-6	1.2206e-6	10,904.5	6,528.26
18	Nominal	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7066e-4	4.9101e-6	1.3174e-6	9,821.7	6,389.34
19	Nominal	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4056e-4	5.2712e-6	1.4143e-6	8,821.7	6,219.56
20	Nominal	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1804e-3	5.9932e-6	1.6081e-6	7,088.2	5,840.56
21	Nominal	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7202e-3	6.7153e-6	1.8018e-6	5,650.3	5,433.51
22	Nominal	station-21	$\epsilon_s/\epsilon_y = 4$	-2.7998e-3	8.1595e-6	2.1893e-6	3,317.6	4,638.20
23	Nominal	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8794e-3	9.6036e-6	2.5768e-6	1,691.1	3,959.54

#	Stage	Station	Criterion	ε0	κx (1/mm)	κy (1/mm)	P (kN)	M0 (kN·m)
24	Nominal	station-23	εs/εy = 7.5	-6.5784e-3	1.3214e-5	3.5455e-6	-1,007.8	2,687.69
25	Nominal	station-24	εs/εy = 10	-9.2774e-3	1.6824e-5	4.5142e-6	-2,707.5	1,811.36
26	Nominal	station-25	εs/εy = 20	-2.0073e-2	3.1266e-5	8.3890e-6	-4,988.5	515.58
27	Nominal	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-5,790.6	0.00

Fixed-P curve points — Mx-My contour at Pu = 24,942.9 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed. The Design surface does not reach this axial force, so it contributes no rows and is absent from the diagram.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Nominal	-180.000	-3,973.95	-0.00	3,973.95
2	Nominal	-174.366	-3,904.69	-385.20	3,923.64
3	Nominal	-165.722	-3,833.05	-975.44	3,955.22
4	Nominal	-164.449	-3,810.21	-1,060.35	3,955.01
5	Nominal	-158.042	-3,690.04	-1,487.74	3,978.66
6	Nominal	-153.637	-3,592.34	-1,780.36	4,009.31
7	Nominal	-152.796	-3,574.90	-1,837.58	4,019.53
8	Nominal	-143.199	-3,300.06	-2,468.88	4,121.38
9	Nominal	-141.587	-3,248.43	-2,575.85	4,145.76
10	Nominal	-132.794	-2,899.05	-3,131.30	4,267.26
11	Nominal	-131.336	-2,834.54	-3,222.43	4,291.70
12	Nominal	-124.804	-2,469.04	-3,551.90	4,325.76
13	Nominal	-122.040	-2,322.29	-3,710.73	4,377.51
14	Nominal	-116.364	-1,960.31	-3,955.30	4,414.44
15	Nominal	-113.389	-1,769.55	-4,091.37	4,457.65
16	Nominal	-107.783	-1,379.81	-4,302.02	4,517.88
17	Nominal	-105.227	-1,197.26	-4,398.33	4,558.37
18	Nominal	-99.125	-736.41	-4,584.80	4,643.57
19	Nominal	-97.550	-613.62	-4,629.98	4,670.47
20	Nominal	-90.159	-13.31	-4,809.96	4,809.98
21	Nominal	-90.000	-0.01	-4,812.16	4,812.16
22	Nominal	-83.802	504.67	-4,646.98	4,674.31
23	Nominal	-82.451	613.62	-4,629.98	4,670.46
24	Nominal	-77.013	1,028.60	-4,460.13	4,577.20
25	Nominal	-74.773	1,197.26	-4,398.32	4,558.36
26	Nominal	-69.371	1,578.16	-4,192.27	4,479.48
27	Nominal	-66.611	1,769.55	-4,091.36	4,457.64
28	Nominal	-60.788	2,145.40	-3,836.80	4,395.88
29	Nominal	-57.961	2,322.27	-3,710.72	4,377.48
30	Nominal	-50.273	2,751.40	-3,310.91	4,304.92
31	Nominal	-48.664	2,834.52	-3,222.42	4,291.68
32	Nominal	-40.139	3,176.08	-2,678.23	4,154.56
33	Nominal	-38.413	3,248.44	-2,575.86	4,145.78
34	Nominal	-28.209	3,544.40	-1,901.18	4,022.09
35	Nominal	-27.204	3,574.90	-1,837.58	4,019.53
36	Nominal	-21.958	3,690.04	-1,487.74	3,978.66
37	Nominal	-14.278	3,833.05	-975.44	3,955.22
38	Nominal	-6.967	3,907.81	-477.56	3,936.89
39	Nominal	0.000	3,973.95	0.00	3,973.95
40	Nominal	5.634	3,904.69	385.20	3,923.64
41	Nominal	14.278	3,833.05	975.45	3,955.22
42	Nominal	15.551	3,810.21	1,060.35	3,955.01
43	Nominal	21.958	3,690.04	1,487.76	3,978.67
44	Nominal	26.363	3,592.34	1,780.37	4,009.32
45	Nominal	27.204	3,574.90	1,837.58	4,019.53
46	Nominal	36.801	3,300.06	2,468.88	4,121.38
47	Nominal	38.413	3,248.43	2,575.85	4,145.76

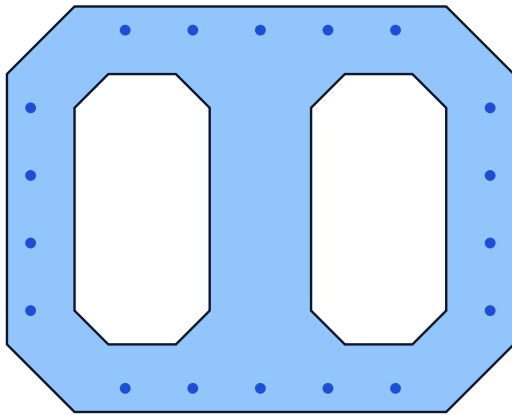
#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
48	Nominal	47.206	2,899.05	3,131.30	4,267.26
49	Nominal	48.664	2,834.54	3,222.43	4,291.70
50	Nominal	55.196	2,469.04	3,551.90	4,325.76
51	Nominal	57.960	2,322.29	3,710.73	4,377.51
52	Nominal	63.636	1,960.31	3,955.30	4,414.44
53	Nominal	66.611	1,769.55	4,091.37	4,457.65
54	Nominal	72.217	1,379.81	4,302.02	4,517.88
55	Nominal	74.773	1,197.26	4,398.33	4,558.37
56	Nominal	80.875	736.41	4,584.80	4,643.57
57	Nominal	82.450	613.62	4,629.98	4,670.47
58	Nominal	89.842	13.30	4,809.97	4,809.98
59	Nominal	90.000	-0.00	4,812.17	4,812.17
60	Nominal	96.198	-504.68	4,646.98	4,674.31
61	Nominal	97.549	-613.62	4,629.98	4,670.47
62	Nominal	102.987	-1,028.60	4,460.13	4,577.20
63	Nominal	105.227	-1,197.26	4,398.32	4,558.36
64	Nominal	110.629	-1,578.16	4,192.27	4,479.48
65	Nominal	113.389	-1,769.55	4,091.36	4,457.64
66	Nominal	119.212	-2,145.40	3,836.80	4,395.88
67	Nominal	122.039	-2,322.27	3,710.72	4,377.48
68	Nominal	129.727	-2,751.40	3,310.91	4,304.92
69	Nominal	131.336	-2,834.52	3,222.42	4,291.68
70	Nominal	139.861	-3,176.08	2,678.23	4,154.56
71	Nominal	141.587	-3,248.44	2,575.86	4,145.78
72	Nominal	151.791	-3,544.39	1,901.17	4,022.09
73	Nominal	152.796	-3,574.90	1,837.57	4,019.53
74	Nominal	158.042	-3,690.05	1,487.72	3,978.66
75	Nominal	165.722	-3,833.05	975.44	3,955.22
76	Nominal	173.033	-3,907.81	477.57	3,936.89
77	Nominal	-180.000	-3,973.95	-0.00	3,973.95

5.2 ULS-2 same axial, half moment (inside)

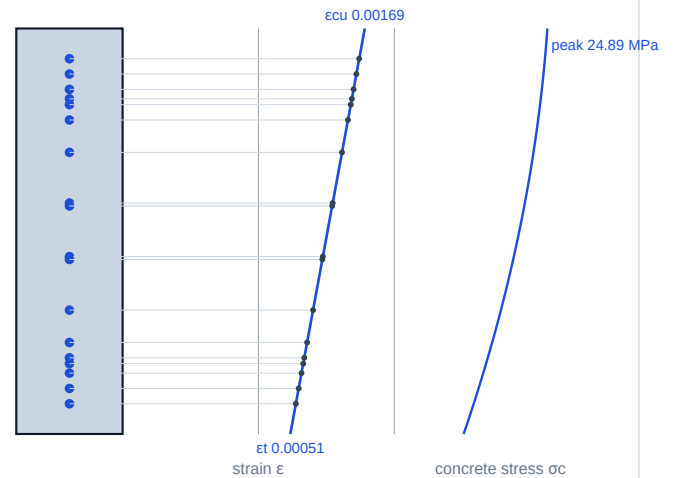
Worked section state and the points behind both curves. $P_u = 24942.9 \text{ kN}$ · $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$ · $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta=1456$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Concrete force **22,045.7 kN**
 Concrete M_y **563.3 kN·m**
 Displaced concrete **-282.9 kN**
 Response P **24,942.9 kN**
 Response M_y **715.9 kN·m**
 Residual $|\Delta M_x|$ **5.19e-3 N·mm**

Concrete M_x **1,476.8 kN·m**
 Steel force (gross) **3,180.1 kN**
 Steel net **2,897.2 kN**
 Response M_x **1,857.1 kN·m**
 Residual $|\Delta P|$ **1.43e-5 N**
 Residual $|\Delta M_y|$ **1.92e-3 N·mm**

$c = 2079.0$

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	328.6	1.423e-3	284.64	228.92	—
2	-200.0	530.0	804.2	273.6	1.468e-3	293.59	236.12	—
3	0.0	530.0	804.2	218.6	1.513e-3	302.54	243.32	—
4	200.0	530.0	804.2	163.6	1.557e-3	311.49	250.51	—
5	400.0	530.0	804.2	108.6	1.602e-3	320.43	257.71	—
6	-400.0	-530.0	804.2	1347.7	5.946e-4	118.92	95.64	—
7	-200.0	-530.0	804.2	1292.7	6.393e-4	127.87	102.84	—
8	0.0	-530.0	804.2	1237.7	6.841e-4	136.81	110.03	—
9	200.0	-530.0	804.2	1182.7	7.288e-4	145.76	117.23	—
10	400.0	-530.0	804.2	1127.7	7.735e-4	154.71	124.42	—
11	-680.0	300.0	804.2	626.8	1.181e-3	236.16	189.93	—
12	-680.0	100.0	804.2	819.1	1.024e-3	204.89	164.78	—
13	-680.0	-100.0	804.2	1011.4	8.681e-4	173.62	139.63	—
14	-680.0	-300.0	804.2	1203.7	7.118e-4	142.35	114.49	—
15	680.0	300.0	804.2	252.7	1.485e-3	297.00	238.86	—
16	680.0	100.0	804.2	444.9	1.329e-3	265.73	213.71	—
17	680.0	-100.0	804.2	637.2	1.172e-3	234.46	188.57	—
18	680.0	-300.0	804.2	829.5	1.016e-3	203.19	163.42	—

Solver and admissibility evidence

Strain plane ε_0 **1.0984e-3**
 Curvature k_y **2.2369e-7**
 Compression depth c **2,079.010 mm**
 Relative residual **3.635e-13**
 Concrete limit **0.003300**

Curvature k_x **7.8173e-7**
 Neutral-axis line **164.0318°**
 Iterations **5**
 Max concrete compression **0.001690**
 Max steel tension **0.000000**

Steel limit

not declared

Strain admissibility

admissible

Governing utilization

1.4116

Vertical curve points — direct meridian at $\beta_{eq} = 15.968^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN-m)
1	Design	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	17,670.3	0.00
2	Design	station-1	$c/D = 3$	2.7500e-3	7.2619e-7	2.0779e-7	17,670.3	335.03
3	Design	station-2	$c/D = 2$	2.4750e-3	1.0893e-6	3.1169e-7	17,670.3	670.07
4	Design	station-3	$c/D = 1.5$	2.2000e-3	1.4524e-6	4.1559e-7	17,670.3	1,005.10
5	Design	station-4	$c/D = 1.2$	1.9250e-3	1.8155e-6	5.1949e-7	17,670.3	1,340.14
6	Design	station-5	$c/D = 1.1$	1.8000e-3	1.9805e-6	5.6671e-7	17,670.3	1,675.17
7	Design	station-6	$c/D = 1$	1.6500e-3	2.1786e-6	6.2338e-7	17,599.1	2,022.30
8	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.5171e-3	2.3541e-6	6.7360e-7	16,245.9	2,565.72
9	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2469e-3	2.7107e-6	7.7566e-7	13,739.1	3,397.40
10	Design	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1119e-3	2.8891e-6	8.2669e-7	12,711.5	3,656.08
11	Design	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7681e-4	3.0674e-6	8.7772e-7	11,815.7	3,840.95
12	Design	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4174e-4	3.2457e-6	9.2875e-7	11,027.5	3,975.38
13	Design	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0667e-4	3.4241e-6	9.7978e-7	10,304.5	4,079.66
14	Design	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7160e-4	3.6024e-6	1.0308e-6	9,633.5	4,162.32
15	Design	station-14	$\epsilon_s/\epsilon_y = 1$	4.3654e-4	3.7808e-6	1.0818e-6	9,004.8	4,229.15
16	Design	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6640e-4	4.1374e-6	1.1839e-6	8,316.6	4,496.44
17	Design	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0374e-4	4.4941e-6	1.2860e-6	7,766.8	4,669.16
18	Design	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7388e-4	4.8508e-6	1.3880e-6	7,296.1	4,778.20
19	Design	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4402e-4	5.2075e-6	1.4901e-6	6,835.3	4,853.94
20	Design	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1843e-3	5.9208e-6	1.6942e-6	5,918.3	4,933.96
21	Design	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7246e-3	6.6341e-6	1.8983e-6	4,678.4	4,581.03
22	Design	station-21	$\epsilon_s/\epsilon_y = 4$	-2.8051e-3	8.0608e-6	2.3066e-6	2,670.7	3,894.64
23	Design	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8857e-3	9.4875e-6	2.7148e-6	1,316.6	3,320.89
24	Design	station-23	$\epsilon_s/\epsilon_y = 7.5$	-6.5871e-3	1.3054e-5	3.7354e-6	-955.8	2,245.97
25	Design	station-24	$\epsilon_s/\epsilon_y = 10$	-9.2884e-3	1.6621e-5	4.7560e-6	-2,364.4	1,514.52
26	Design	station-25	$\epsilon_s/\epsilon_y = 20$	-2.0094e-2	3.0888e-5	8.8384e-6	-4,270.2	423.69
27	Design	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-4,922.0	0.00
1	Nominal	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	33,981.4	-0.00
2	Nominal	station-1	$c/D = 3$	2.7500e-3	7.2619e-7	2.0779e-7	33,981.4	-0.00
3	Nominal	station-2	$c/D = 2$	2.4750e-3	1.0893e-6	3.1169e-7	33,854.0	72.79
4	Nominal	station-3	$c/D = 1.5$	2.2000e-3	1.4524e-6	4.1559e-7	32,790.6	610.95
5	Nominal	station-4	$c/D = 1.2$	1.9250e-3	1.8155e-6	5.1949e-7	30,564.8	1,625.29
6	Nominal	station-5	$c/D = 1.1$	1.8000e-3	1.9805e-6	5.6671e-7	29,150.1	2,243.82
7	Nominal	station-6	$c/D = 1$	1.6500e-3	2.1786e-6	6.2338e-7	27,075.5	3,111.23
8	Nominal	station-7	$\epsilon_s/\epsilon_y = 0$	1.5171e-3	2.3541e-6	6.7360e-7	24,993.7	3,947.26
9	Nominal	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2469e-3	2.7107e-6	7.7566e-7	21,137.0	5,226.77
10	Nominal	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1119e-3	2.8891e-6	8.2669e-7	19,556.1	5,624.74
11	Nominal	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7681e-4	3.0674e-6	8.7772e-7	18,178.0	5,909.15
12	Nominal	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4174e-4	3.2457e-6	9.2875e-7	16,965.4	6,115.98
13	Nominal	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0667e-4	3.4241e-6	9.7978e-7	15,853.1	6,276.39
14	Nominal	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7160e-4	3.6024e-6	1.0308e-6	14,820.8	6,403.57
15	Nominal	station-14	$\epsilon_s/\epsilon_y = 1$	4.3654e-4	3.7808e-6	1.0818e-6	13,853.6	6,506.39
16	Nominal	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6640e-4	4.1374e-6	1.1839e-6	12,170.7	6,580.15
17	Nominal	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0374e-4	4.4941e-6	1.2860e-6	10,837.4	6,515.11
18	Nominal	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7388e-4	4.8508e-6	1.3880e-6	9,728.1	6,370.94
19	Nominal	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4402e-4	5.2075e-6	1.4901e-6	8,725.9	6,196.52
20	Nominal	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1843e-3	5.9208e-6	1.6942e-6	6,962.7	5,804.66
21	Nominal	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7246e-3	6.6341e-6	1.8983e-6	5,504.1	5,389.45
22	Nominal	station-21	$\epsilon_s/\epsilon_y = 4$	-2.8051e-3	8.0608e-6	2.3066e-6	3,142.0	4,581.93
23	Nominal	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8857e-3	9.4875e-6	2.7148e-6	1,548.9	3,906.93

#	Stage	Station	Criterion	ε0	κx (1/mm)	κy (1/mm)	P (kN)	M0 (kN·m)
24	Nominal	station-23	εs/εy = 7.5	-6.5871e-3	1.3054e-5	3.7354e-6	-1,124.5	2,642.32
25	Nominal	station-24	εs/εy = 10	-9.2884e-3	1.6621e-5	4.7560e-6	-2,781.6	1,781.78
26	Nominal	station-25	εs/εy = 20	-2.0094e-2	3.0888e-5	8.8384e-6	-5,023.8	498.46
27	Nominal	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-5,790.6	0.00

Fixed-P curve points — Mx-My contour at Pu = 24,942.9 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed. The Design surface does not reach this axial force, so it contributes no rows and is absent from the diagram.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Nominal	-180.000	-3,973.95	-0.00	3,973.95
2	Nominal	-174.366	-3,904.69	-385.20	3,923.64
3	Nominal	-165.722	-3,833.05	-975.44	3,955.22
4	Nominal	-164.449	-3,810.21	-1,060.35	3,955.01
5	Nominal	-158.042	-3,690.04	-1,487.74	3,978.66
6	Nominal	-153.637	-3,592.34	-1,780.36	4,009.31
7	Nominal	-152.796	-3,574.90	-1,837.58	4,019.53
8	Nominal	-143.199	-3,300.06	-2,468.88	4,121.38
9	Nominal	-141.587	-3,248.43	-2,575.85	4,145.76
10	Nominal	-132.794	-2,899.05	-3,131.30	4,267.26
11	Nominal	-131.336	-2,834.54	-3,222.43	4,291.70
12	Nominal	-124.804	-2,469.04	-3,551.90	4,325.76
13	Nominal	-122.040	-2,322.29	-3,710.73	4,377.51
14	Nominal	-116.364	-1,960.31	-3,955.30	4,414.44
15	Nominal	-113.389	-1,769.55	-4,091.37	4,457.65
16	Nominal	-107.783	-1,379.81	-4,302.02	4,517.88
17	Nominal	-105.227	-1,197.26	-4,398.33	4,558.37
18	Nominal	-99.125	-736.41	-4,584.80	4,643.57
19	Nominal	-97.550	-613.62	-4,629.98	4,670.47
20	Nominal	-90.159	-13.31	-4,809.96	4,809.98
21	Nominal	-90.000	-0.01	-4,812.16	4,812.16
22	Nominal	-83.802	504.67	-4,646.98	4,674.31
23	Nominal	-82.451	613.62	-4,629.98	4,670.46
24	Nominal	-77.013	1,028.60	-4,460.13	4,577.20
25	Nominal	-74.773	1,197.26	-4,398.32	4,558.36
26	Nominal	-69.371	1,578.16	-4,192.27	4,479.48
27	Nominal	-66.611	1,769.55	-4,091.36	4,457.64
28	Nominal	-60.788	2,145.40	-3,836.80	4,395.88
29	Nominal	-57.961	2,322.27	-3,710.72	4,377.48
30	Nominal	-50.273	2,751.40	-3,310.91	4,304.92
31	Nominal	-48.664	2,834.52	-3,222.42	4,291.68
32	Nominal	-40.139	3,176.08	-2,678.23	4,154.56
33	Nominal	-38.413	3,248.44	-2,575.86	4,145.78
34	Nominal	-28.209	3,544.40	-1,901.18	4,022.09
35	Nominal	-27.204	3,574.90	-1,837.58	4,019.53
36	Nominal	-21.958	3,690.04	-1,487.74	3,978.66
37	Nominal	-14.278	3,833.05	-975.44	3,955.22
38	Nominal	-6.967	3,907.81	-477.56	3,936.89
39	Nominal	0.000	3,973.95	0.00	3,973.95
40	Nominal	5.634	3,904.69	385.20	3,923.64
41	Nominal	14.278	3,833.05	975.45	3,955.22
42	Nominal	15.551	3,810.21	1,060.35	3,955.01
43	Nominal	21.958	3,690.04	1,487.76	3,978.67
44	Nominal	26.363	3,592.34	1,780.37	4,009.32
45	Nominal	27.204	3,574.90	1,837.58	4,019.53
46	Nominal	36.801	3,300.06	2,468.88	4,121.38
47	Nominal	38.413	3,248.43	2,575.85	4,145.76

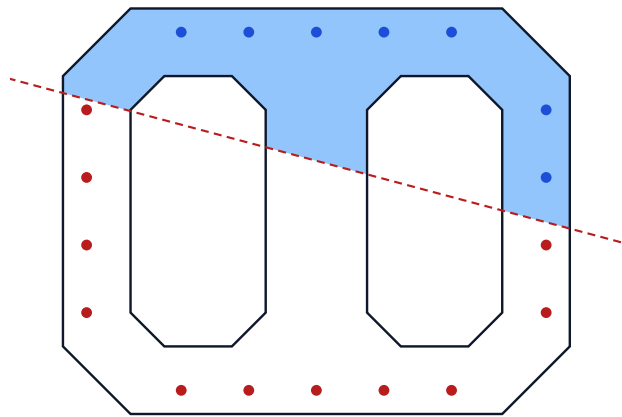
#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
48	Nominal	47.206	2,899.05	3,131.30	4,267.26
49	Nominal	48.664	2,834.54	3,222.43	4,291.70
50	Nominal	55.196	2,469.04	3,551.90	4,325.76
51	Nominal	57.960	2,322.29	3,710.73	4,377.51
52	Nominal	63.636	1,960.31	3,955.30	4,414.44
53	Nominal	66.611	1,769.55	4,091.37	4,457.65
54	Nominal	72.217	1,379.81	4,302.02	4,517.88
55	Nominal	74.773	1,197.26	4,398.33	4,558.37
56	Nominal	80.875	736.41	4,584.80	4,643.57
57	Nominal	82.450	613.62	4,629.98	4,670.47
58	Nominal	89.842	13.30	4,809.97	4,809.98
59	Nominal	90.000	-0.00	4,812.17	4,812.17
60	Nominal	96.198	-504.68	4,646.98	4,674.31
61	Nominal	97.549	-613.62	4,629.98	4,670.47
62	Nominal	102.987	-1,028.60	4,460.13	4,577.20
63	Nominal	105.227	-1,197.26	4,398.32	4,558.36
64	Nominal	110.629	-1,578.16	4,192.27	4,479.48
65	Nominal	113.389	-1,769.55	4,091.36	4,457.64
66	Nominal	119.212	-2,145.40	3,836.80	4,395.88
67	Nominal	122.039	-2,322.27	3,710.72	4,377.48
68	Nominal	129.727	-2,751.40	3,310.91	4,304.92
69	Nominal	131.336	-2,834.52	3,222.42	4,291.68
70	Nominal	139.861	-3,176.08	2,678.23	4,154.56
71	Nominal	141.587	-3,248.44	2,575.86	4,145.78
72	Nominal	151.791	-3,544.39	1,901.17	4,022.09
73	Nominal	152.796	-3,574.90	1,837.57	4,019.53
74	Nominal	158.042	-3,690.05	1,487.72	3,978.66
75	Nominal	165.722	-3,833.05	975.44	3,955.22
76	Nominal	173.033	-3,907.81	477.57	3,936.89
77	Nominal	-180.000	-3,973.95	-0.00	3,973.95

5.3 ULS-3 low axial, high moment

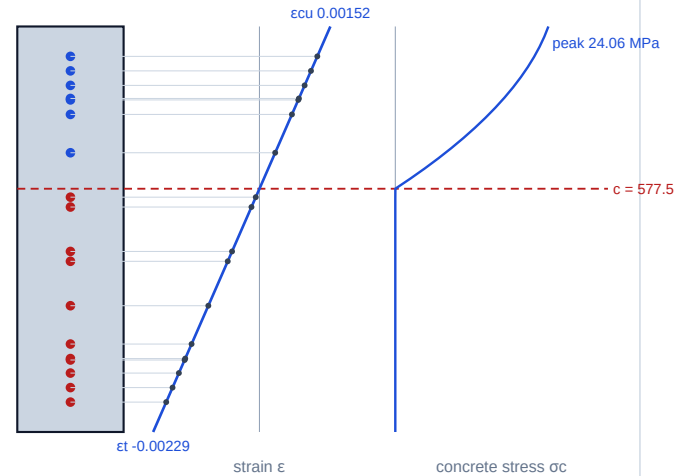
Worked section state and the points behind both curves. $P_u = 5000.0 \text{ kN}$ · $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 20.56^\circ$

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta=1443$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Concrete force	6,195.6 kN	Concrete Mx	2,728.2 kN·m
Concrete My	1,023.3 kN·m	Steel force (gross)	-1,101.1 kN
Displaced concrete	-94.5 kN	Steel net	-1,195.6 kN
Response P	5,000.0 kN	Response Mx	4,000.0 kN·m
Response My	1,500.0 kN·m	Residual $ \Delta P $	5.59e-9 N
Residual $ \Delta M_x $	0.00e+0 N·mm	Residual $ \Delta M_y $	9.54e-7 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.0	6.983e-4	139.66	112.33	—
2	-200.0	530.0	804.2	261.4	8.347e-4	166.94	134.26	—
3	0.0	530.0	804.2	209.7	9.711e-4	194.22	156.20	—
4	200.0	530.0	804.2	158.0	1.108e-3	221.50	178.14	—
5	400.0	530.0	804.2	106.4	1.244e-3	248.78	200.08	—
6	-400.0	-530.0	804.2	1337.1	-2.005e-3	-400.00	-321.70	—
7	-200.0	-530.0	804.2	1285.4	-1.869e-3	-373.74	-300.58	—
8	0.0	-530.0	804.2	1233.7	-1.732e-3	-346.46	-278.64	—
9	200.0	-530.0	804.2	1182.1	-1.596e-3	-319.18	-256.70	—
10	400.0	-530.0	804.2	1130.4	-1.460e-3	-291.91	-234.76	—
11	-680.0	300.0	804.2	607.6	-7.922e-5	-15.84	-12.74	—
12	-680.0	100.0	804.2	800.8	-5.893e-4	-117.86	-94.79	—
13	-680.0	-100.0	804.2	994.0	-1.099e-3	-219.88	-176.84	—
14	-680.0	-300.0	804.2	1187.2	-1.609e-3	-321.89	-258.88	—
15	680.0	300.0	804.2	256.2	8.483e-4	169.65	136.44	—
16	680.0	100.0	804.2	449.4	3.382e-4	67.64	54.40	—
17	680.0	-100.0	804.2	642.7	-1.719e-4	-34.38	-27.65	—
18	680.0	-300.0	804.2	835.9	-6.820e-4	-136.40	-109.70	—

Solver and admissibility evidence

Strain plane ε_0	-3.8060e-4	Curvature k_x	2.5504e-6
Curvature k_y	6.8197e-7	Neutral-axis line	165.0295°
Compression depth c	577.545 mm	Iterations	6
Relative residual	1.419e-16	Max concrete compression	0.001525
Concrete limit	0.003300	Max steel tension	0.002005

Steel limit

not declared

Strain admissibility

admissible

Governing utilization

0.8774

Vertical curve points — direct meridian at $\beta_{eq} = 14.971^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN·m)
1	Design	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	17,670.3	0.00
2	Design	station-1	$c/D = 3$	2.7500e-3	7.3621e-7	1.9686e-7	17,670.3	336.03
3	Design	station-2	$c/D = 2$	2.4750e-3	1.1043e-6	2.9529e-7	17,670.3	672.05
4	Design	station-3	$c/D = 1.5$	2.2000e-3	1.4724e-6	3.9372e-7	17,670.3	1,008.08
5	Design	station-4	$c/D = 1.2$	1.9250e-3	1.8405e-6	4.9215e-7	17,670.3	1,344.11
6	Design	station-5	$c/D = 1.1$	1.8000e-3	2.0078e-6	5.3689e-7	17,670.3	1,680.14
7	Design	station-6	$c/D = 1$	1.6500e-3	2.2086e-6	5.9058e-7	17,556.5	2,035.58
8	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.5187e-3	2.3843e-6	6.3757e-7	16,210.8	2,575.41
9	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2488e-3	2.7456e-6	7.3417e-7	13,700.2	3,409.90
10	Design	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1139e-3	2.9262e-6	7.8247e-7	12,681.6	3,664.72
11	Design	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7895e-4	3.1069e-6	8.3077e-7	11,799.3	3,846.56
12	Design	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4401e-4	3.2875e-6	8.7907e-7	11,020.5	3,979.75
13	Design	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0906e-4	3.4681e-6	9.2737e-7	10,304.7	4,083.84
14	Design	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7412e-4	3.6488e-6	9.7567e-7	9,640.3	4,166.76
15	Design	station-14	$\epsilon_s/\epsilon_y = 1$	4.3917e-4	3.8294e-6	1.0240e-6	9,017.0	4,234.28
16	Design	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6928e-4	4.1907e-6	1.1206e-6	8,350.5	4,504.43
17	Design	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0061e-4	4.5519e-6	1.2172e-6	7,817.5	4,679.02
18	Design	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7050e-4	4.9132e-6	1.3138e-6	7,369.7	4,792.66
19	Design	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4038e-4	5.2745e-6	1.4104e-6	6,914.2	4,872.95
20	Design	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1802e-3	5.9970e-6	1.6036e-6	6,030.5	4,966.09
21	Design	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7199e-3	6.7195e-6	1.7968e-6	4,809.2	4,620.46
22	Design	station-21	$\epsilon_s/\epsilon_y = 4$	-2.7995e-3	8.1646e-6	2.1832e-6	2,827.8	3,945.03
23	Design	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8791e-3	9.6096e-6	2.5696e-6	1,443.9	3,368.04
24	Design	station-23	$\epsilon_s/\epsilon_y = 7.5$	-6.5780e-3	1.3222e-5	3.5356e-6	-851.2	2,286.67
25	Design	station-24	$\epsilon_s/\epsilon_y = 10$	-9.2768e-3	1.6835e-5	4.5016e-6	-2,298.0	1,541.06
26	Design	station-25	$\epsilon_s/\epsilon_y = 20$	-2.0072e-2	3.1285e-5	8.3657e-6	-4,238.6	439.04
27	Design	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-4,922.0	0.00
1	Nominal	pure-compression	Pure compression	3.3000e-3	0.0000e+0	0.0000e+0	33,981.4	-0.00
2	Nominal	station-1	$c/D = 3$	2.7500e-3	7.3621e-7	1.9686e-7	33,981.4	-0.00
3	Nominal	station-2	$c/D = 2$	2.4750e-3	1.1043e-6	2.9529e-7	33,850.4	74.33
4	Nominal	station-3	$c/D = 1.5$	2.2000e-3	1.4724e-6	3.9372e-7	32,774.1	616.71
5	Nominal	station-4	$c/D = 1.2$	1.9250e-3	1.8405e-6	4.9215e-7	30,526.3	1,637.30
6	Nominal	station-5	$c/D = 1.1$	1.8000e-3	2.0078e-6	5.3689e-7	29,095.9	2,259.00
7	Nominal	station-6	$c/D = 1$	1.6500e-3	2.2086e-6	5.9058e-7	27,010.0	3,131.66
8	Nominal	station-7	$\epsilon_s/\epsilon_y = 0$	1.5187e-3	2.3843e-6	6.3757e-7	24,939.7	3,962.17
9	Nominal	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.2488e-3	2.7456e-6	7.3417e-7	21,077.3	5,246.01
10	Nominal	station-9	$\epsilon_s/\epsilon_y = 0.375$	1.1139e-3	2.9262e-6	7.8247e-7	19,510.2	5,638.02
11	Nominal	station-10	$\epsilon_s/\epsilon_y = 0.5$	9.7895e-4	3.1069e-6	8.3077e-7	18,152.7	5,917.79
12	Nominal	station-11	$\epsilon_s/\epsilon_y = 0.625$	8.4401e-4	3.2875e-6	8.7907e-7	16,954.6	6,122.69
13	Nominal	station-12	$\epsilon_s/\epsilon_y = 0.75$	7.0906e-4	3.4681e-6	9.2737e-7	15,853.3	6,282.83
14	Nominal	station-13	$\epsilon_s/\epsilon_y = 0.875$	5.7412e-4	3.6488e-6	9.7567e-7	14,831.2	6,410.39
15	Nominal	station-14	$\epsilon_s/\epsilon_y = 1$	4.3917e-4	3.8294e-6	1.0240e-6	13,872.3	6,514.27
16	Nominal	station-15	$\epsilon_s/\epsilon_y = 1.25$	1.6928e-4	4.1907e-6	1.1206e-6	12,220.3	6,591.85
17	Nominal	station-16	$\epsilon_s/\epsilon_y = 1.5$	-1.0061e-4	4.5519e-6	1.2172e-6	10,908.2	6,528.86
18	Nominal	station-17	$\epsilon_s/\epsilon_y = 1.75$	-3.7050e-4	4.9132e-6	1.3138e-6	9,826.2	6,390.21
19	Nominal	station-18	$\epsilon_s/\epsilon_y = 2$	-6.4038e-4	5.2745e-6	1.4104e-6	8,826.6	6,220.78
20	Nominal	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.1802e-3	5.9970e-6	1.6036e-6	7,094.7	5,842.46
21	Nominal	station-20	$\epsilon_s/\epsilon_y = 3$	-1.7199e-3	6.7195e-6	1.7968e-6	5,657.9	5,435.83
22	Nominal	station-21	$\epsilon_s/\epsilon_y = 4$	-2.7995e-3	8.1646e-6	2.1832e-6	3,326.8	4,641.22
23	Nominal	station-22	$\epsilon_s/\epsilon_y = 5$	-3.8791e-3	9.6096e-6	2.5696e-6	1,698.7	3,962.40

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M θ (kN·m)
24	Nominal	station-23	$\epsilon_s/\epsilon_y = 7.5$	-6.5780e-3	1.3222e-5	3.5356e-6	-1,001.5	2,690.20
25	Nominal	station-24	$\epsilon_s/\epsilon_y = 10$	-9.2768e-3	1.6835e-5	4.5016e-6	-2,703.5	1,813.01
26	Nominal	station-25	$\epsilon_s/\epsilon_y = 20$	-2.0072e-2	3.1285e-5	8.3657e-6	-4,986.6	516.51
27	Nominal	pure-tension	Pure tension	-5.0000e-2	0.0000e+0	0.0000e+0	-5,790.6	0.00

Fixed-P curve points — Mx-My contour at Pu = 5,000.0 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Design	-180.000	-4,724.11	-0.01	4,724.11
2	Design	-177.542	-4,691.17	-201.41	4,695.50
3	Design	-166.253	-4,548.50	-1,112.79	4,682.64
4	Design	-161.586	-4,434.57	-1,476.35	4,673.86
5	Design	-153.000	-4,203.75	-2,141.93	4,717.99
6	Design	-146.252	-3,953.10	-2,641.13	4,754.21
7	Design	-141.305	-3,687.11	-2,953.38	4,724.12
8	Design	-135.796	-3,390.86	-3,297.94	4,730.15
9	Design	-130.955	-3,125.05	-3,600.66	4,767.67
10	Design	-124.938	-2,775.86	-3,973.48	4,847.05
11	Design	-121.067	-2,556.55	-4,243.62	4,954.22
12	Design	-117.120	-2,279.58	-4,450.80	5,000.61
13	Design	-112.138	-1,932.24	-4,749.58	5,127.57
14	Design	-109.848	-1,754.96	-4,861.89	5,168.93
15	Design	-103.824	-1,288.65	-5,236.79	5,393.02
16	Design	-103.682	-1,276.14	-5,242.11	5,395.20
17	Design	-103.399	-1,252.45	-5,257.60	5,404.72
18	Design	-98.325	-788.92	-5,391.15	5,448.57
19	Design	-96.579	-629.27	-5,455.79	5,491.96
20	Design	-93.606	-346.27	-5,494.18	5,505.08
21	Design	-90.000	0.00	-5,548.05	5,548.05
22	Design	-88.557	139.23	-5,528.41	5,530.16
23	Design	-83.421	629.26	-5,455.79	5,491.96
24	Design	-76.601	1,252.41	-5,257.62	5,404.73
25	Design	-76.176	1,288.65	-5,236.78	5,393.00
26	Design	-75.940	1,308.28	-5,223.74	5,385.07
27	Design	-67.862	1,932.23	-4,749.59	5,127.59
28	Design	-64.567	2,172.23	-4,567.83	5,058.03
29	Design	-58.933	2,556.58	-4,243.63	4,954.24
30	Design	-53.267	2,893.66	-3,877.49	4,838.20
31	Design	-49.045	3,125.03	-3,600.63	4,767.64
32	Design	-43.480	3,425.33	-3,248.20	4,720.56
33	Design	-38.695	3,687.11	-2,953.42	4,724.14
34	Design	-36.512	3,792.55	-2,807.52	4,718.65
35	Design	-33.749	3,953.04	-2,641.24	4,754.23
36	Design	-29.563	4,101.37	-2,326.44	4,715.24
37	Design	-27.000	4,203.74	-2,141.96	4,717.99
38	Design	-22.124	4,319.58	-1,756.07	4,662.89
39	Design	-18.413	4,434.58	-1,476.32	4,673.87
40	Design	-15.389	4,496.67	-1,237.70	4,663.90
41	Design	-13.748	4,548.50	-1,112.82	4,682.65
42	Design	-7.759	4,586.06	-624.85	4,628.43
43	Design	-2.458	4,691.18	-201.41	4,695.50
44	Design	-1.292	4,698.69	-105.99	4,699.89
45	Design	0.000	4,724.11	0.01	4,724.11
46	Design	2.458	4,691.17	201.41	4,695.50
47	Design	13.747	4,548.50	1,112.79	4,682.64
48	Design	18.414	4,434.57	1,476.35	4,673.86

#	Stage	θM (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
49	Design	27.000	4,203.75	2,141.95	4,717.99
50	Design	33.747	3,953.10	2,641.12	4,754.21
51	Design	38.695	3,687.12	2,953.38	4,724.12
52	Design	44.204	3,390.86	3,297.93	4,730.15
53	Design	49.045	3,125.06	3,600.66	4,767.67
54	Design	55.062	2,775.86	3,973.48	4,847.05
55	Design	58.934	2,556.54	4,243.64	4,954.23
56	Design	62.880	2,279.58	4,450.80	5,000.61
57	Design	67.863	1,932.22	4,749.60	5,127.59
58	Design	70.152	1,754.96	4,861.90	5,168.94
59	Design	76.176	1,288.65	5,236.79	5,393.02
60	Design	76.318	1,276.14	5,242.11	5,395.20
61	Design	76.601	1,252.45	5,257.60	5,404.72
62	Design	81.675	788.92	5,391.15	5,448.57
63	Design	83.421	629.27	5,455.79	5,491.96
64	Design	86.394	346.27	5,494.18	5,505.08
65	Design	90.000	-0.00	5,548.05	5,548.05
66	Design	91.443	-139.23	5,528.41	5,530.16
67	Design	96.579	-629.26	5,455.79	5,491.96
68	Design	103.399	-1,252.41	5,257.62	5,404.73
69	Design	103.824	-1,288.65	5,236.78	5,393.00
70	Design	104.060	-1,308.28	5,223.74	5,385.07
71	Design	112.137	-1,932.23	4,749.60	5,127.59
72	Design	115.433	-2,172.21	4,567.84	5,058.03
73	Design	121.067	-2,556.58	4,243.62	4,954.24
74	Design	126.733	-2,893.67	3,877.47	4,838.19
75	Design	130.955	-3,125.03	3,600.63	4,767.64
76	Design	136.520	-3,425.33	3,248.20	4,720.56
77	Design	141.305	-3,687.11	2,953.42	4,724.14
78	Design	143.488	-3,792.55	2,807.52	4,718.65
79	Design	146.251	-3,953.04	2,641.24	4,754.23
80	Design	150.437	-4,101.37	2,326.44	4,715.24
81	Design	153.000	-4,203.74	2,141.95	4,717.99
82	Design	157.876	-4,319.58	1,756.07	4,662.89
83	Design	161.587	-4,434.58	1,476.32	4,673.87
84	Design	164.611	-4,496.67	1,237.70	4,663.90
85	Design	166.252	-4,548.50	1,112.82	4,682.65
86	Design	172.241	-4,586.06	624.85	4,628.43
87	Design	177.542	-4,691.18	201.41	4,695.50
88	Design	178.708	-4,698.69	105.99	4,699.89
89	Design	-180.000	-4,724.11	-0.01	4,724.11
1	Nominal	-180.000	-5,265.03	-0.04	5,265.03
2	Nominal	-179.840	-5,263.00	-14.70	5,263.03
3	Nominal	-179.442	-5,258.15	-51.21	5,258.40
4	Nominal	-170.479	-5,125.93	-859.72	5,197.52
5	Nominal	-165.399	-5,054.40	-1,316.63	5,223.07
6	Nominal	-152.931	-4,712.33	-2,408.18	5,292.01
7	Nominal	-152.117	-4,687.41	-2,480.05	5,303.06
8	Nominal	-150.022	-4,600.28	-2,653.57	5,310.75
9	Nominal	-140.070	-4,166.43	-3,487.34	5,433.29
10	Nominal	-130.772	-3,659.21	-4,243.50	5,603.31
11	Nominal	-130.411	-3,637.99	-4,273.03	5,611.94
12	Nominal	-122.746	-3,092.16	-4,808.06	5,716.55
13	Nominal	-121.129	-2,979.98	-4,934.41	5,764.43
14	Nominal	-115.014	-2,473.99	-5,302.12	5,850.91
15	Nominal	-112.263	-2,245.39	-5,484.80	5,926.62

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
16	Nominal	-108.161	-1,867.02	-5,691.49	5,989.90
17	Nominal	-104.053	-1,479.18	-5,909.44	6,091.75
18	Nominal	-102.730	-1,343.82	-5,948.44	6,098.34
19	Nominal	-100.377	-1,106.84	-6,044.39	6,144.90
20	Nominal	-97.546	-812.46	-6,133.53	6,187.11
21	Nominal	-96.706	-725.88	-6,173.64	6,216.17
22	Nominal	-92.552	-276.82	-6,212.16	6,218.33
23	Nominal	-90.000	-0.00	-6,266.37	6,266.37
24	Nominal	-85.640	473.68	-6,212.97	6,231.01
25	Nominal	-83.294	725.90	-6,173.64	6,216.17
26	Nominal	-79.622	1,106.95	-6,044.35	6,144.88
27	Nominal	-75.947	1,479.17	-5,909.45	6,091.76
28	Nominal	-73.114	1,748.91	-5,761.24	6,020.85
29	Nominal	-67.737	2,245.35	-5,484.83	5,926.63
30	Nominal	-61.454	2,772.90	-5,097.20	5,802.62
31	Nominal	-58.872	2,979.95	-4,934.42	5,764.43
32	Nominal	-49.953	3,613.85	-4,299.64	5,616.66
33	Nominal	-49.590	3,638.00	-4,273.06	5,611.96
34	Nominal	-42.222	4,037.55	-3,663.89	5,452.15
35	Nominal	-39.930	4,166.39	-3,487.39	5,433.30
36	Nominal	-34.706	4,380.17	-3,033.67	5,328.13
37	Nominal	-27.882	4,687.43	-2,479.99	5,303.05
38	Nominal	-27.420	4,699.94	-2,438.33	5,294.80
39	Nominal	-27.068	4,712.35	-2,408.13	5,292.01
40	Nominal	-19.051	4,898.79	-1,691.62	5,182.63
41	Nominal	-14.601	5,054.40	-1,316.64	5,223.07
42	Nominal	-11.931	5,073.23	-1,071.98	5,185.25
43	Nominal	-9.521	5,125.93	-859.68	5,197.52
44	Nominal	-5.012	5,158.01	-452.37	5,177.81
45	Nominal	-0.159	5,263.00	-14.64	5,263.02
46	Nominal	-0.061	5,263.17	-5.61	5,263.17
47	Nominal	0.000	5,265.03	0.04	5,265.03
48	Nominal	0.160	5,263.00	14.72	5,263.02
49	Nominal	0.559	5,258.14	51.31	5,258.39
50	Nominal	9.521	5,125.93	859.72	5,197.52
51	Nominal	14.601	5,054.40	1,316.62	5,223.07
52	Nominal	27.069	4,712.33	2,408.18	5,292.01
53	Nominal	27.883	4,687.41	2,480.07	5,303.06
54	Nominal	29.978	4,600.27	2,653.61	5,310.75
55	Nominal	39.930	4,166.43	3,487.34	5,433.29
56	Nominal	49.228	3,659.21	4,243.50	5,603.31
57	Nominal	49.589	3,637.99	4,273.03	5,611.94
58	Nominal	57.254	3,092.15	4,808.06	5,716.54
59	Nominal	58.872	2,979.96	4,934.42	5,764.43
60	Nominal	64.986	2,473.98	5,302.12	5,850.90
61	Nominal	67.737	2,245.36	5,484.82	5,926.62
62	Nominal	71.839	1,867.02	5,691.50	5,989.90
63	Nominal	75.947	1,479.18	5,909.44	6,091.75
64	Nominal	77.270	1,343.82	5,948.44	6,098.34
65	Nominal	79.623	1,106.84	6,044.39	6,144.90
66	Nominal	82.454	812.46	6,133.53	6,187.11
67	Nominal	83.294	725.88	6,173.64	6,216.17
68	Nominal	87.448	276.82	6,212.16	6,218.33
69	Nominal	90.000	-0.00	6,266.37	6,266.37
70	Nominal	94.360	-473.68	6,212.97	6,231.01
71	Nominal	96.706	-725.90	6,173.64	6,216.17

#	Stage	θ_M (°)	M_x (kN·m)	M_y (kN·m)	$ M $ (kN·m)
72	Nominal	100.378	-1,106.95	6,044.35	6,144.88
73	Nominal	104.053	-1,479.17	5,909.45	6,091.76
74	Nominal	106.886	-1,748.90	5,761.25	6,020.85
75	Nominal	112.263	-2,245.34	5,484.83	5,926.63
76	Nominal	118.547	-2,772.92	5,097.18	5,802.62
77	Nominal	121.128	-2,979.96	4,934.41	5,764.43
78	Nominal	130.047	-3,613.85	4,299.64	5,616.66
79	Nominal	130.410	-3,638.00	4,273.06	5,611.96
80	Nominal	137.778	-4,037.55	3,663.89	5,452.15
81	Nominal	140.070	-4,166.39	3,487.39	5,433.30
82	Nominal	145.294	-4,380.17	3,033.67	5,328.13
83	Nominal	152.118	-4,687.43	2,479.98	5,303.05
84	Nominal	152.580	-4,699.94	2,438.33	5,294.80
85	Nominal	152.932	-4,712.35	2,408.13	5,292.01
86	Nominal	160.949	-4,898.79	1,691.62	5,182.63
87	Nominal	165.399	-5,054.40	1,316.63	5,223.07
88	Nominal	168.069	-5,073.23	1,071.98	5,185.25
89	Nominal	170.479	-5,125.93	859.68	5,197.52
90	Nominal	174.988	-5,158.01	452.37	5,177.81
91	Nominal	179.841	-5,263.01	14.61	5,263.03
92	Nominal	179.939	-5,263.17	5.60	5,263.18
93	Nominal	-180.000	-5,265.03	-0.04	5,265.03